



Features of special number sequences

Task A: Types of special number sequences

1) Fill in the missing numbers in these special number sequences.

triangular numbers	square numbers	cube numbers
1, 3, __, 10, __, 21, 28, __, ...	__, 4, 9, __, 25, 36, __, __, ...	1, __, 27, __, __, 216, ...

2) Calculate the following:

- The 5th square number plus the 5th **triangular number** ...
- The 8th **triangular number** plus the 4th cube number ...
- The 1st **triangular number** plus the 1st cube number ...
- The 3rd square number minus the 2nd cube number ...
- The 10th cube number minus the 10th square number ...
- The 11th **triangular number** plus the 12th **triangular number** ...

3) Sofia is having fun building different sequences using the square, triangular and cube numbers. Suggest what the next number in each sequence could be, explaining your answer.

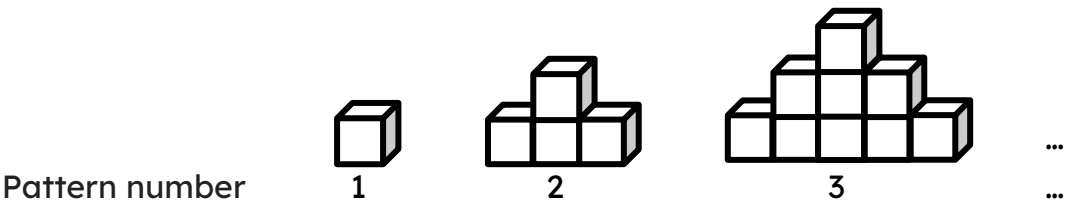
- 81, 64, 49, 36, ...
- 1, 6, 15, 28, ...
- 2, 7, 15, 26, 40, ...
- 0, 4, 18, 48, 100, ...
- 1, 1, 3, 4, 6, 9, 10, 16, ...
- 0, 1, 3, 6, 10, 15, 21, ...

Some of these were built from **more than one** type of number. Writing out the start of the sequence of square, **triangular** and cube numbers may help you see what I did.



Task B: Identifying special number sequences

1) Izzy is building step patterns out of cubes. Here are her first 3 patterns.



a) Fill in the table to show the number of cubes in her first 5 patterns if she continues to build them in the same way.

Pattern number	1	2	3	4	5
Number of cubes					

- b) What do you notice about your answers?
- c) How many cubes would she need to build the 10th pattern?

2) Alex is playing around with number patterns. He notices that if he squares the number 3, then subtracts one, then divides by 8 he gets a square number. He wants to see if this works for other odd numbers.

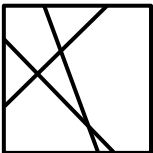
a) Fill in the table with the results from the first 5 odd numbers greater than 1

Start number (<i>n</i>)	3	5	7	9	11
Result ($\frac{n^2 - 1}{8}$)					

- b) What do you notice about your answers?
- c) What value would you get if you applied this calculation to the number 19?

3) Sam is investigating the maximum number of intersection points between straight lines drawn in a square. They want to know how many intersection points there could be if they drew 12 different lines, but drawing them all is proving difficult.

a) Sam tries it with 3 lines first, making sure the lines intersect the maximum number of times. How many intersection points are there?



b) Fill in the table for 2 lines, 3 lines, 4 lines and 5 lines, what do you notice about your answers?

Number of lines	2	3	4	5
Number of intersections				

c) What is the maximum number of intersections with 12 lines?